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THEORETICAL AND EXPERIMENTAL DYNAMIC ANALYSIS OF THE H. B. ROBINSON NUCLEAR PLANT

REACTORS

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A mathematical model for predicting the dynamic response of the H. B. Robinson pressurized water reactor plant was formulated and compared with results from measurements made during full-power operation of the plant. The model was based on the basic conservation laws for neutrons, mass, and energy; design data from the safety analysis report were used to evaluate the necessary coefficients. The model included representations for point kinetics, core heat transfer, piping, pressurizer, and the steam generator. The experiment involved perturbations in control rod position and main steam valve opening. Periodic binary input signals and step inputs were used. Theoretical and experimental frequency responses were obtained from the model and the test data. The comparison showed that the model was capable of good predictions for reactivity perturbations and fair predictions for steam valve perturbations. A method was also demonstrated for using the test data for at-power determination of the differential control rod worth.

I. INTRODUCTION

Dynamic response measurements were made at the H. B. Robinson Nuclear Plant. The experimental results were correlated with theoretical predictions to check the validity of a mathematical model of the plant developed for system simulation. Frequency response and step response tests were performed using control rods and the main steam valve for introducing reactivity and steam flow perturbations. The theoretical model is a linear state-variable model that includes the re-

actor core, pressurizer, primary system piping, and a U-tube recirculation-type steam generator. The test program provided data to check the validity of the theoretical model and to evaluate the practicality of small perturbation dynamic testing for routine performance evaluation in power reactors.

II. DESCRIPTION OF THE SYSTEM

The H. B. Robinson Nuclear Plant¹ produces 2200 MW(th) [739 MW(e)] at full power. It includes a pressurized water reactor (PWR), pressurizer, and three vertical U-tube recirculation-type steam generators.

Normal plant equipment was used to provide test perturbations, and normal plant sensors were used to measure system responses. Figure 1 is a schematic that shows plant equipment used in the test program. Magnetic jack control rod drive mechanisms are used. The control rods move in a step-wise manner, with each step $\frac{5}{8}$ in. in magnitude and with a minimum time between steps of 0.833 sec.

The system outputs were monitored at the junction box where the signals enter the plant computer.

III. THEORETICAL SYSTEM ANALYSIS

III.A. Introduction

An objective of this program was to contribute to the development of an experimentally confirmed theoretical model for PWR systems. The modeling effort in this program involved the use of state variable methods. This formulation consists of a set of coupled first-order differential equations. This is preferred for modeling large complex

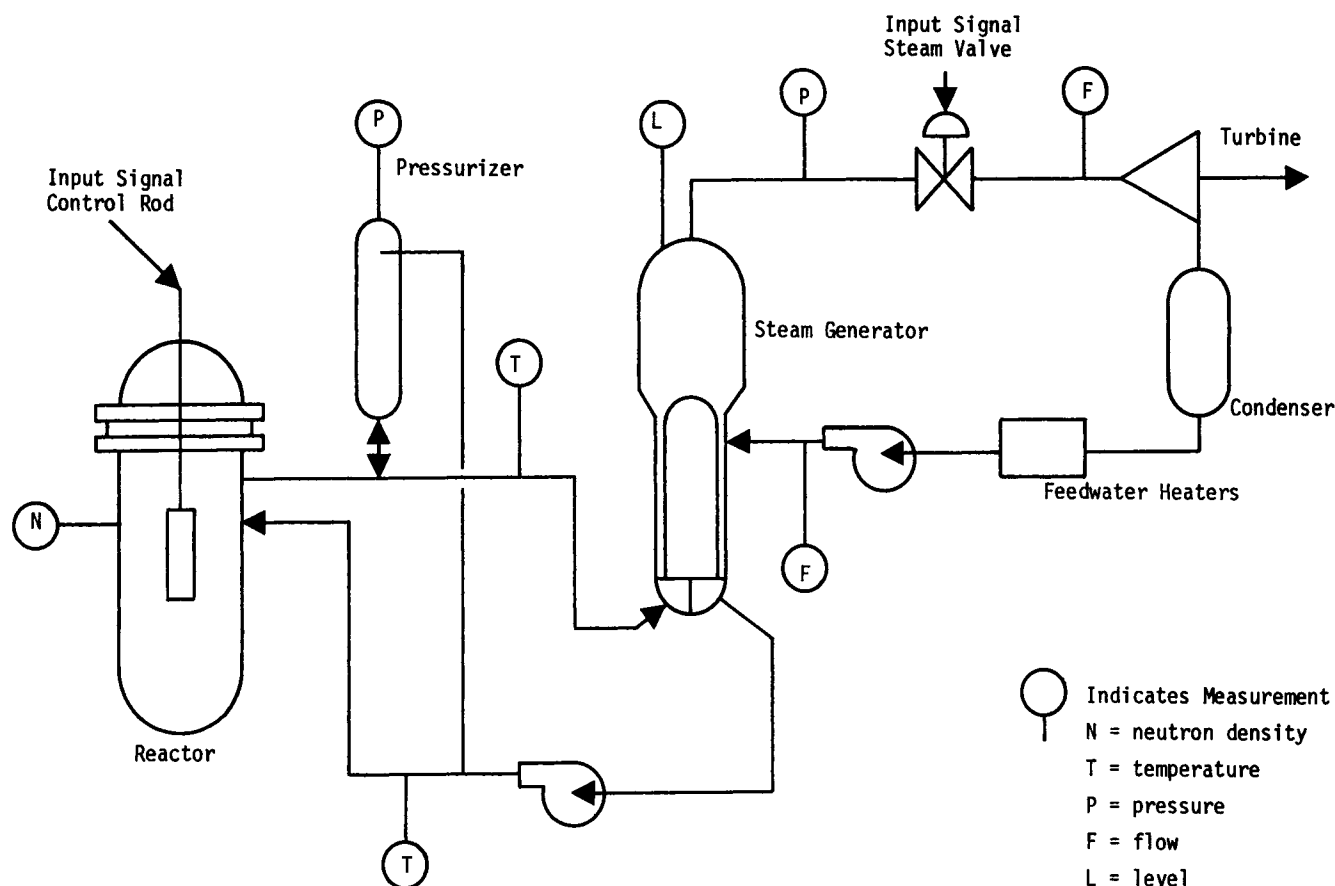


Fig. 1. Schematic of the H. B. Robinson Nuclear Plant.

systems such as nuclear power plants because of the following advantages:

1. Standard general-purpose computer codes can be used for the analyses, eliminating the need for code development each time an analysis is to be performed for a different system.
2. The coefficient matrix can be used for stability analysis, transient response analysis, and frequency response analysis without any model reformulation.
3. Subsystems can be modeled and checked individually, then easily coupled with the other subsystems to give a complete system model.
4. Changes in system coefficients or modeling assumptions are easy to make.
5. Addition of nonlinear effects to a linear model to give a nonlinear model is straightforward.
6. Control system models are easily included for control system synthesis work.

III.B. The Model

The model for the H. B. Robinson Nuclear Plant was developed, and numerical values for the coefficients were evaluated using the data in Tables I, II, and III. The equations are described below.

III.B.1. Core

The reactor power was modeled using the point kinetics equations with six groups of delayed neutrons and reactivity feedbacks due to changes in fuel temperature, coolant temperature, and primary coolant system pressure. The linearized equations are:

$$\begin{aligned}
 \frac{d\delta P}{dt} = & -\frac{\beta}{\Lambda} \delta P + \sum_i \lambda_i \delta C_i + \frac{\alpha_f P_0}{\Lambda} \sum_{\text{fuel nodes}} F_{fi} \delta T_{fi} \\
 & + \frac{\alpha_p P_0}{\Lambda} \delta P_p + \frac{P_0}{\Lambda} \delta \rho_{\text{rod}} \\
 & + \frac{\alpha_c P_0}{\Lambda} \sum_{\text{coolant nodes}} F_{ci} \delta T_{ci}
 \end{aligned} \quad (1)$$

TABLE I
Reactor Design Data

Core Thermal and Hydraulic Characteristics (Ref. 1)			
1. Total primary heat output, MW(th)	2200 (7508 × 10 ⁶ Btu/h)		
2. Nominal primary system pressure, psi	2250		
3. Total coolant flow rate, lb/h	101.5 × 10 ⁶		
4. Average coolant velocity along fuel rods, ft/sec	14.3		
5. Total mass of coolant in primary loop, lb	406 050		
6. Nominal coolant inlet temperature, °F	546.2		
7. Nominal coolant outlet temperature, °F	602.1		
8. Active heat transfer surface area, ft ²	42 460		
9. Average heat flux, Btu/(h ft ²)	171 600		
10. Fuel-to-coolant heat transfer coefficient (includes resistance in fuel) Btu/(h ft ² °F)	176		
Kinetic Characteristics			
11. Doppler coefficient (Δk/k)/°F	-1.3 × 10 ⁻⁵		
12. Moderator temperature coefficient, (Δk/k)/°F	-2.0 × 10 ⁻⁴		
13. Moderator pressure coefficient, (Δk/k)/psi	+3.0 × 10 ⁻⁶		
14. Prompt neutron lifetime, sec	1.6 × 10 ⁻⁵		
15. Delayed neutron fraction	0.0064		
16. Delayed neutron constants:			
	Mean Life (sec)	Decay Constant (λ _i , sec ⁻¹)	Fraction
	80.4	0.0124	0.00021
	32.8	0.0305	0.00140
	8.98	0.111	0.00125
	3.32	0.301	0.00253
	0.88	1.14	0.00074
	0.332	3.01	0.00027

$$\frac{d\delta C_i}{dt} = \frac{\beta_i}{\Lambda} \delta P - \lambda_i \delta C_i \quad , \quad (2)$$

where

δP = deviation of reactor power from initial steady-state value

P_0 = initial steady-state power level

λ_i = delayed neutron decay constant for i 'th delayed neutron group

δC_i = deviation of normalized precursor concentration from its steady-state value

P_p = primary system pressure

α_f = fuel temperature coefficient of reactivity

α_c = coolant temperature coefficient of reactivity

α_p = coolant pressure coefficient of reactivity

δT_{fi} = deviation of fuel temperature in the i 'th fuel node from its initial steady-state value

δT_{ci} = deviation of coolant temperature in the i 'th coolant node from its initial steady-state value

$\delta \rho_{rod}$ = reactivity due to control rod movement

β_i = delayed neutron fraction for the i 'th delayed neutron group

β = total delayed neutron fraction

TABLE II
Steam Generator Data
(for each unit)

Steam Generator	
1. Number of U-tubes	3260
2. U-tube diameter, in.	0.875
3. Average tube wall thickness, in.	0.050
4. Mass of U-tube metal, lb	91 800
5. Total heat transfer area, ft ²	44 430
Steam Conditions at Full Load	
6. Steam flow, lb/h	3.169×10^6
7. Steam temperature, °F	516
8. Steam pressure, psig	770
Primary Side Coolant	
9. Reactor coolant flow, lb/h	33.93×10^6
10. Reactor coolant water volume, ft ³	928
Secondary Side Fluid	
11. Feedwater temperature, °F	435
12. Secondary side water volume, full power, ft ³	1526
13. Secondary side steam volume, full power, ft ³	3203

TABLE III
Pressurizer Design Data

1. Water volume, full power, ft ³	780
2. Steam volume, full power, ft ³	520
3. Electric heater capacity, kW (total)	1300

Λ = neutron generation time

F_{fi} = reactivity importance for temperature changes in the i 'th fuel node

F_{ci} = reactivity importance for temperature changes in the i 'th coolant node.

Data from Table I were used to evaluate the coefficients. For a model with one fuel node and two coolant nodes, the resulting equations are (with $F_{fi} = 1$ and $F_{ci} = 0.5$):

$$\begin{aligned} \frac{d\delta P}{dt} = & -400 \delta P + 0.0125 \delta C_1 + 0.0305 \delta C_2 \\ & + 0.111 \delta C_3 + 0.301 \delta C_4 + 1.140 \delta C_5 \\ & + 3.01 \delta C_6 - 1781 \delta T_f - 13\,700 \delta T_{C1} \\ & - 13\,700 \delta T_{C2} + 411 \delta P_p \end{aligned} \quad (3)$$

$$\frac{d\delta C_1}{dt} = 13.125 \delta P - 0.0125 \delta C_1 \quad (4)$$

$$\frac{d\delta C_2}{dt} = 87.5 \delta P - 0.0305 \delta C_2 \quad (5)$$

$$\frac{d\delta C_3}{dt} = 78.125 \delta P - 0.111 \delta C_3 \quad (6)$$

$$\frac{d\delta C_4}{dt} = 158.125 \delta P - 0.301 \delta C_4 \quad (7)$$

$$\frac{d\delta C_5}{dt} = 46.25 \delta P - 1.140 \delta C_5 \quad (8)$$

$$\frac{d\delta C_6}{dt} = 16.875 \delta P - 3.01 \delta C_6 \quad (9)$$

The core heat transfer model used a nodal approximation for fuel and coolant temperatures. Each axial section included a fuel temperature node and two coolant temperature nodes as shown schematically in Fig. 2. Two coolant nodes are used for each fuel node to obtain a good approximation to the average coolant temperature. This modeling approach is used in favor of other possibilities, such as the well-mixed approximation (outlet temperature = average temperature) or the arithmetic average approximation (average temperature = $\frac{1}{2}$ inlet temperature + $\frac{1}{2}$ outlet temperature). In the model with two coolant nodes for each fuel node, each coolant node is based on a well-mixed approximation. The average coolant temperature used to establish the temperature difference for heat flux determination is the first-node temperature. This provides a better representation than the well-mixed or arithmetic average approximation. The equations are:

$$\frac{d\delta T_{fi}}{dt} = \frac{Q_{fi}}{(MC_p)_{fi}} \delta P - \left(\frac{UA_f}{(MC_p)_{fi}} \right) (\delta T_{fi} - \delta T_{cli}) \quad (10)$$

$$\frac{d\delta T_{cli}}{dt} = \left(\frac{UA_f}{(MC_p)_{ci}} \right) (\delta T_{fi} - \delta T_{cli}) - \frac{2}{\tau} (\delta T_{cli} - \delta T_{cin}) \quad (11)$$

$$\frac{d\delta T_{c2i}}{dt} = \left(\frac{UA_f}{(MC_p)_{ci}} \right) (\delta T_{fi} - \delta T_{cli}) - \frac{2}{\tau} (\delta T_{c2i} - \delta T_{cli}) \quad (12)$$

where

δT_{fi} = average fuel temperature

δT_{cli} = average coolant temperature in i 'th fuel node

δT_{c2i} = outlet coolant temperature in i 'th fuel node

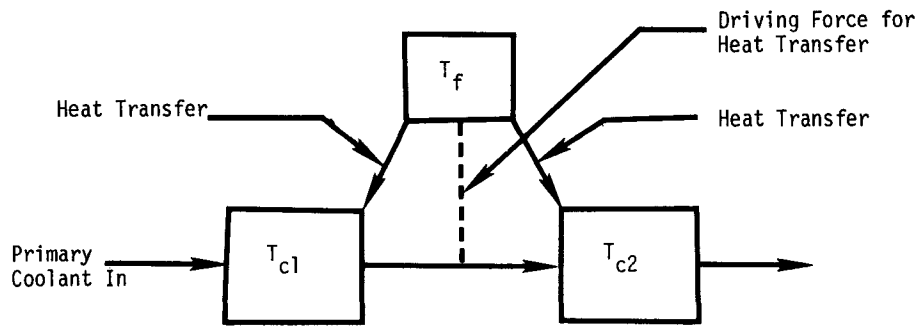


Fig. 2. Schematic of the fuel-coolant heat transfer model.

Q_{fi} = fraction of total reactor power generated in fuel node i

$(MC_p)_{fi}$ = total heat capacity for i 'th fuel node

$(MC_p)_{ci}$ = total heat capacity of both coolant nodes associated with i 'th fuel node

U = overall fuel-to-coolant heat transfer coefficient (includes resistance in fuel as well as film resistance)

A_f = heat transfer area

τ = residence time (both coolant nodes)

δT_{cin} = deviation in inlet temperature of the first coolant node from its initial steady-state value.

Two different core heat transfer models were formulated: a detailed one with 45 heat transfer nodes (15 for fuel and 30 for coolant), and a simplified one with 3 heat transfer nodes (1 for fuel and 2 for coolant). Figure 3 shows a comparison of results for an isolated (constant inlet temperature) reactor core obtained with both models. Because of the good agreement, the simpler model was used for the core in the complete system model.

Evaluation of the coefficients yields

$$\frac{d\delta T_f}{dt} = 0.0756 \delta P - 0.16466 \delta T_f + 0.16466 \delta T_{c1} \quad (13)$$

$$\frac{d\delta T_{c1}}{dt} = 0.05707 \delta T_f - 2.4403 \delta T_{c1} + 2.3832 \delta T_{LP} \quad (14)$$

$$\frac{d\delta T_{c2}}{dt} = 0.05707 \delta T_f + 2.3262 \delta T_{c1} - 2.3832 \delta T_{c2} \quad , \quad (15)$$

where δT_{LP} is the reactor lower plenum temperature.

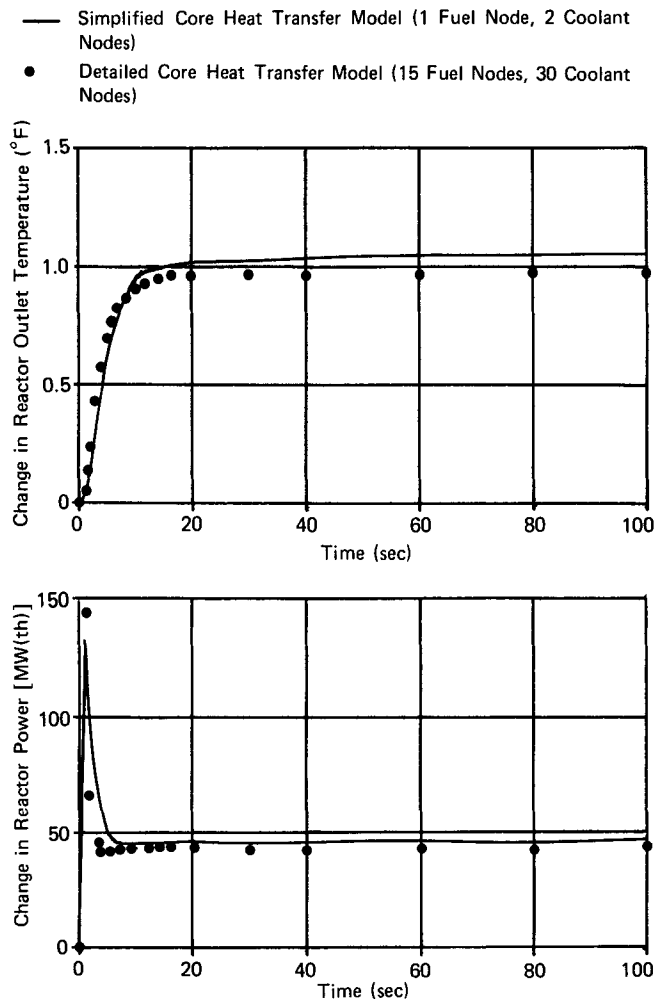


Fig. 3. A comparison of results for simplified and detailed core models for a 7.1¢ step change in reactivity.

III.B.2. Pressurizer

The pressurizer model is based on mass, energy, and volume balances and the assumption that saturation conditions always apply for the steam-water mixture in the pressurizer.

The basic equations are:

1. water mass balance

$$\frac{dM_w}{dt} = W_{wi} - W_s \quad (16)$$

2. steam mass balance

$$\frac{dM_s}{dt} = W_s \quad (17)$$

3. water energy balance

$$\frac{dE_w}{dt} = W_{wi} h_{wi} - W_s h_s - P \dot{V}_w + q \quad (18)$$

4. volume balance

$$V_w + V_s = V_t \quad (19)$$

5. compressibility-corrected perfect gas law

$$P_p V_s = M_s R T_s, \quad (20)$$

where

M_w = mass of water in the pressurizer

M_s = mass of steam in the pressurizer

W_{wi} = mass flow of water into (or out of) the pressurizer

W_s = flashing rate (or condensing rate) in the pressurizer

E_w = internal energy of water in the pressurizer

h_{wi} = enthalpy of water entering the pressurizer

h_s = enthalpy of steam in the pressurizer

P_p = pressure in the pressurizer

q = rate of heat addition to the pressurizer fluid with electric heater

R = gas constant

T_s = saturation temperature

V_w = volume of water in the pressurizer

V_s = volume of steam in the pressurizer.

The equations are linearized and manipulated (see Ref. 2) to obtain

$$\frac{d\delta P_p}{dt} = B_1 \delta P_p + B_2 \delta W_w + B_3 \delta q \quad (21)$$

The values of B_1 , B_2 , and B_3 for the H. R. Robinson Nuclear Plant are:

$$B_1 = -1.913 \times 10^{-6} \text{ (sec}^{-1}\text{)}$$

$$B_2 = 7.021 \times 10^{-3} \text{ (psi/lb)}$$

$$B_3 = 2.1726 \times 10^{-4} \text{ [psi/(kW sec)]}$$

The change in mass in the pressurizer is obtained by summing the contribution due to expansion (or contraction) of the water in each coolant node in the primary loop as follows:

$$\delta M_w = \sum_{i=1}^N V_i \gamma_i \delta T_{ci} \quad (22)$$

or

$$\delta W_w = \sum_{i=1}^N V_i \gamma_i \frac{d\delta T_{ci}}{dt}, \quad (23)$$

where

V_i = volume of i 'th coolant node

γ_i = slope of coolant density versus temperature curve

T_{ci} = temperature of i 'th coolant node.

Evaluation of the coefficients for the H. B. Robinson Nuclear Plant gives

$$\begin{aligned} \delta W_w = & 85.33 \frac{d\delta T_{LP}}{dt} + 25.83 \frac{d\delta T_{c1}}{dt} + 25.83 \frac{d\delta T_{c2}}{dt} \\ & + 187.5 \frac{d\delta T_{up}}{dt} + 39.2 \frac{d\delta T_{HL}}{dt} + 39.54 \frac{d\delta T_{IP}}{dt} \\ & + 171.55 \frac{d\delta T_p}{dt} + 39.54 \frac{d\delta T_{OP}}{dt} + 27.44 \frac{d\delta T_{CL}}{dt}, \end{aligned} \quad (24)$$

where

δT_{LP} = reactor lower plenum temperature

δT_{c1} = coolant temperature in node 1

δT_{c2} = coolant temperature in node 2

δT_{up} = reactor upper plenum temperature

δT_{HL} = hot-leg temperature

δT_{IP} = temperature of primary coolant in the steam generator inlet plenum

δT_p = temperature of primary coolant node in the steam generator

δT_{OP} = temperature of primary coolant in the steam generator outlet plenum

δT_{CL} = cold-leg temperature.

The original model had no representation of the pressurizer control system. However, as is shown in Sec. VI, the test results showed that the pressurizer controller must be included in the model to avoid discrepancies between theory and experiment. It was not convenient to obtain controller parameters for the H. B. Robinson Nuclear Plant controller, but data were available for Sequoyah (a later-generation Westinghouse PWR). The H. B. Robinson Nuclear Plant model was

revised to include the pressurizer controller, and the Sequoyah controller parameters were used. Of course, this does not represent the true situation at the H. B. Robinson Nuclear Plant, but it permits an assessment of the effect of a pressurizer controller on the correlation between theory and experiment.

The pressurizer controller uses a heater to compensate for steady-state heat losses from the pressurizer. It is also used for pressure control for normal pressure variations. Heat input increases for low pressure and decreases for high pressure. When the pressure goes below the control range of these heaters, auxiliary heaters turn on. When the pressure goes above the control range, spray flow is used to decrease the pressure. The model used in this work includes only the normally operating heater.

The block diagram for the pressure controller is shown in Fig. 4. The transfer functions were used to formulate differential equations for inclusion in the state variable model. The resulting equations were:

$$\begin{aligned} \frac{d\delta P_p}{dt} = & 0.0207 \delta T_f - 0.0207 \delta T_{c1} + 0.0103 \delta T_{c2} \\ & + 0.240 \delta T_{up} - 0.130 \delta T_{LP} - 0.509 \delta T_p \\ & + 0.634 \delta T_m - 0.116 \delta T_{OP} + 0.121 \delta T_{LP} \\ & - 0.279 \delta T_{HL} + 0.0235 \delta T_{CL} - 0.0106 \delta P_p \\ & - 0.00213 \delta X \end{aligned} \quad (25)$$

$$\frac{d\delta X}{dt} = 0.00556 \delta P_p, \quad (26)$$

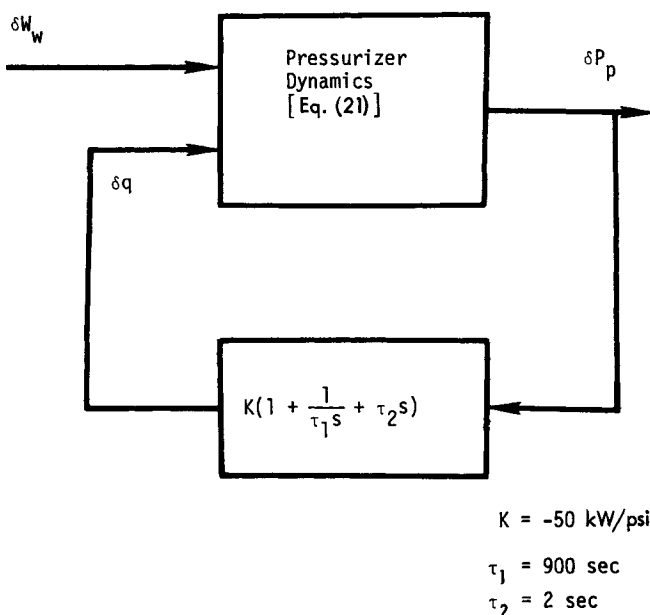


Fig. 4. Pressurizer control system.

where T_m is the steam generator tube metal temperature and X is the integral control action variable. Note that the inclusion of the pressurizer controller affects the matrix by modifying the coefficients in the differential equation for pressurizer pressure and by requiring an additional equation to provide the integral action.

III.B.3. Steam Generator

The steam generator used in this work was a simplified version of a more detailed generator model under development at The University of Tennessee. It uses only three regions to represent the whole steam generator: primary fluid, tube metal, and secondary fluid. The model includes no control action. This is equivalent to assuming that the model applies only for small upsets in which controller dead bands or long-time constants prevent significant changes in the feedwater flow.

The equations are:

1. primary water energy balance

$$\frac{d\delta T_p}{dt} = \frac{1}{\tau_{SG}} \delta T_{IP} - \frac{(hA)_{pm}}{M_p C_p} (\delta T_p - \delta T_m) - \frac{1}{\tau_{SG}} \delta T_p \quad (27)$$

2. metal energy balance

$$\begin{aligned} \frac{d\delta T_m}{dt} = & \frac{(hA)_{pm}}{M_m C_m} (\delta T_p - \delta T_m) \\ & - \frac{(hA)_{ms}}{M_m C_m} \left[\delta T_m - \left(\frac{\partial T_{sat}}{\partial P_s} \right) \delta P_s \right] \end{aligned} \quad (28)$$

3. secondary water (liquid phase) mass balance

$$\frac{d\delta M_w}{dt} = \delta W_{FW} - \delta W_s \quad (29)$$

4. secondary water (steam) mass balance

$$\frac{d\delta M_s}{dt} = \delta W_s - \delta W_{s,0} \quad (30)$$

5. secondary fluid (steam and liquid phase) energy balance

$$\begin{aligned} \frac{d\delta E}{dt} = & (hA)_{ms} \left[\delta T_m - \left(\frac{\partial T_{sat}}{\partial P_s} \right) \delta P_s \right] \\ & + W_{FW} C_{pFW} \left[\delta T_{FW} - \left(\frac{\partial T_{sat}}{\partial P_s} \right) \delta P_s \right] \\ & - h_{st} \delta W_{s,0} \end{aligned} \quad (31)$$

6. equation-of-state

$$P_s V_s = R M_s T_s \quad (32)$$

7. volume balance

$$V_s + V_w = V_T, \quad (33)$$

where

δT_p = primary coolant temperature in the steam generator

τ_{SG} = residence time of coolant in the steam generator

δT_{IP} = primary coolant temperature in the steam generator inlet plenum

h_{pm} = heat transfer coefficient for primary coolant to metal (includes a portion of the metal resistance as well as the film resistance)

h_{st} = enthalpy of steam

A = heat transfer area

δT_m = tube metal temperature

M_p = mass of primary coolant in the steam generator

C_p = specific heat of primary coolant

C_{PFW} = specific heat of feedwater

h_{ms} = heat transfer coefficient for metal to secondary coolant (includes a portion of the metal resistance as well as the film resistance)

M_m = mass of tube metal

C_m = specific heat of tube metal

$\frac{\partial T_{sat}}{\partial P_s}$ = slope of saturation temperature versus saturation pressure curve

P_s = steam pressure

M_w = mass of water in the steam generator

W_{FW} = feedwater flow rate

W_s = steam generation rate

M_s = mass of steam in the steam generator

$W_{s,0}$ = steam flow rate to the turbine

E = internal energy of secondary coolant

V_s = steam volume

T_s = steam temperature

V_w = water volume.

These equations are linearized, and appropriate algebraic substitutions were made to give the following equations:

$$\frac{d\delta T_p}{dt} = \frac{1}{\tau_{SG}} \delta T_{IP} - \frac{(hA)_{pm}}{M_p C_p} (\delta T_p - \delta T_m) - \frac{1}{\tau_{SG}} \delta T_p \quad (34)$$

$$\frac{d\delta T_m}{dt} = \frac{(hA)_{pm}}{M_m C_m} (\delta T_p - \delta T_m) - \frac{(hA)_{ms}}{M_m C_m} \left[\delta T_m - \left(\frac{\delta T_{sat}}{\partial P_s} \right) \delta P_s \right] \quad (35)$$

$$\frac{d\delta P_s}{dt} = D_1 \delta P_s + D_2 \delta T_m + D_3 \delta T_{FW} + D_4 \delta W_{FW} + D_5 \delta W_{s0}, \quad (36)$$

where D_i are the coefficients obtained from algebraic substitutions.

Numerical values for the coefficients were obtained using the H. B. Robinson Nuclear Plant design data. The resulting equations for the steam generator are:

$$\frac{d\delta T_p}{dt} = 0.2238 \delta T_{IP} - 0.76642 \delta T_p + 0.53819 \delta T_m \quad (37)$$

$$\frac{d\delta T_m}{dt} = 3.07017 \delta T_p - 5.3657 \delta T_m + 0.33272 \delta P_s \quad (38)$$

$$\begin{aligned} \frac{d\delta P_s}{dt} = & 1.349 \delta T_m - 0.2034 \delta P_s + 0.05328 \delta T_{FW} \\ & - 0.03843 \delta W_{FW} \\ & - 0.04425 \delta W_{s0}. \end{aligned} \quad (39)$$

Figure 5 shows the calculated response of an isolated steam generator to perturbations in the main steam valve opening.

III.B.4. Piping and Plenums

All piping sections and plenums were modeled as well-mixed volumes. The equation is

$$\frac{d\delta T}{dt} = \frac{1}{\tau} \delta T_{in} - \frac{1}{\tau} \delta T, \quad (40)$$

where

T = temperature of fluid in the section (equal to outlet temperature)

T_{in} = fluid temperature at entrance

τ = fluid residence time.

There were two piping sections (hot-leg and cold-leg) and four plenums (reactor upper, reactor lower, steam generator inlet, and steam generator outlet) in the model.

Substitution of numerical values for each of these yields:

$$\frac{dT_{UP}}{dt} = 0.33645 \delta T_{C2} - 0.33645 \delta T_{UP} \quad (41)$$

$$\frac{dT_{HL}}{dt} = 2.5 \delta T_{UP} - 2.5 \delta T_{HL} \quad (42)$$

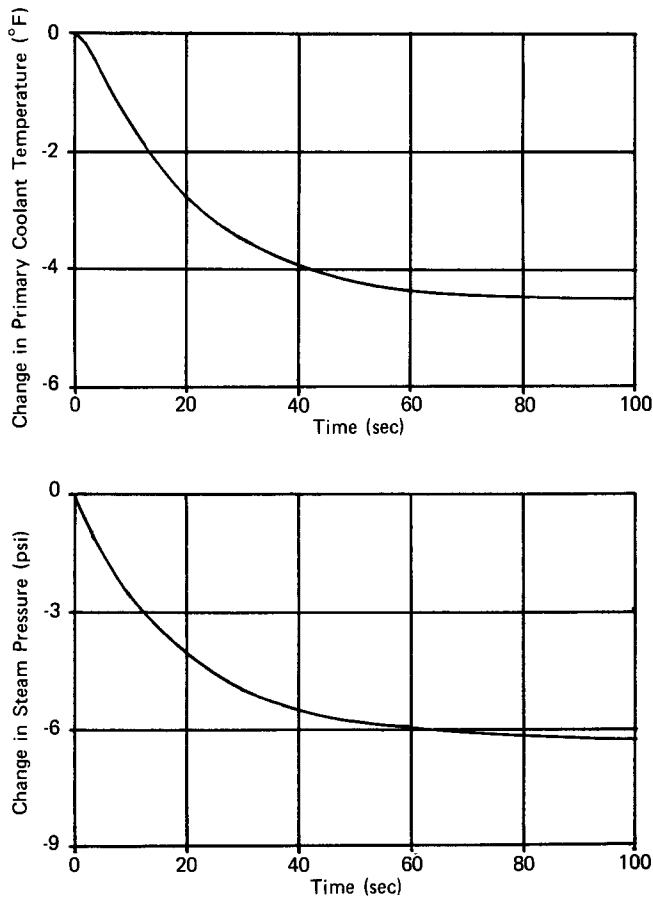


Fig. 5. Response of an isolated steam generator for a 1% increase in steam flow.

$$\frac{dT_{IP}}{dt} = 1.45 \delta T_{HL} - 1.45 \delta T_{IP} \quad (43)$$

$$\frac{dT_{OP}}{dt} = 1.45 \delta T_p - 1.45 \delta T_{OP} \quad (44)$$

$$\frac{dT_{CL}}{dt} = 1.48 \delta T_{OP} - 1.48 \delta T_{CL} \quad (45)$$

$$\frac{dT_{LP}}{dt} = 0.516 \delta T_{CL} - 0.516 \delta T_{LP} \quad (46)$$

III.B.5. Overall System

The nodal structure for the complete system is shown in Fig. 6.

III.C. Methods of Analysis

The transient responses of the isolated subsystems and the complete system were computed using the MATEXP code.³ The system of equations for MATEXP must be in the form:

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x} + \mathbf{g}(\mathbf{x}, t) + \mathbf{f}(t) \quad (47)$$

where

$\mathbf{x}$ = the solution vector

$\mathbf{A}$ = matrix of constant coefficient, linear terms

$\mathbf{g}(\mathbf{x}, t)$ = vector of nonlinear and variable coefficient terms

$\mathbf{f}(t)$ = vector of forcing terms (inputs).

MATEXP uses a matrix-exponential-type of solution. The output at time, $t + \Delta t$, is given by

$$\mathbf{x}(t + \Delta t) = \exp(\mathbf{A}\Delta t) \mathbf{x}(t) + \int_t^{t+\Delta t} \exp[\mathbf{A}(t - t')] \times [\mathbf{g}(\mathbf{x}, t') + \mathbf{f}(t')] dt' \quad (48)$$

If $\mathbf{g}(\mathbf{x}, \tau)$ and $\mathbf{f}(\tau)$ can be assumed to be piecewise constant, the integral can be evaluated, and Eq. (48) becomes

$$\mathbf{x}(t + \Delta t) = \exp(\mathbf{A}\Delta t) \mathbf{x}(t) + [\exp(\mathbf{A}\Delta t) - \mathbf{I}] \mathbf{A}^{-1} \times \{\mathbf{g}[\mathbf{x}(t), t] + \mathbf{f}(t)\} \quad (49)$$

The terms involving matrix exponentials are evaluated as follows:

$$\exp(\mathbf{A}\Delta t) = \mathbf{I} + \mathbf{A}\Delta t + \frac{1}{2!} (\mathbf{A}\Delta t)^2 + \frac{1}{3!} (\mathbf{A}\Delta t)^3 + \dots \quad (50)$$

$$[\exp(\mathbf{A}\Delta t) - \mathbf{I}] \mathbf{A}^{-1} = \Delta t \left[\mathbf{I} + \frac{1}{2!} \mathbf{A}\Delta t + \frac{1}{3!} (\mathbf{A}\Delta t)^2 + \frac{1}{4!} (\mathbf{A}\Delta t)^3 \right] + \dots \quad (51)$$

where $\mathbf{I}$ is the unit diagonal matrix. Note that the second expansion avoids the need for a matrix inversion.

Equation (49) applies for linear systems if $\mathbf{g}(\mathbf{x}, t)$ is set equal to zero and $\mathbf{f}(t)$ is specified. The procedure for MATEXP to handle nonlinear terms is to add a subroutine to evaluate $\mathbf{g}(\mathbf{x}, t)$ at each time step using current values of $\mathbf{x}$. Thus, linear systems require no special programming at all. The analysis of nonlinear systems makes maximum use of a standard solution procedure and requires only the programming needed to furnish the vector of nonlinear terms.

The frequency response was calculated using a computer code called SFR-III (Ref. 4). For a linear constant coefficient model with a single forcing function, Eq. (47) can be rewritten as

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x} + \mathbf{h}q \quad (52)$$

where $\mathbf{h}$ is the vector of constant coefficients of the forcing term and q is the time varying forcing term (a scalar). The vector of frequency responses

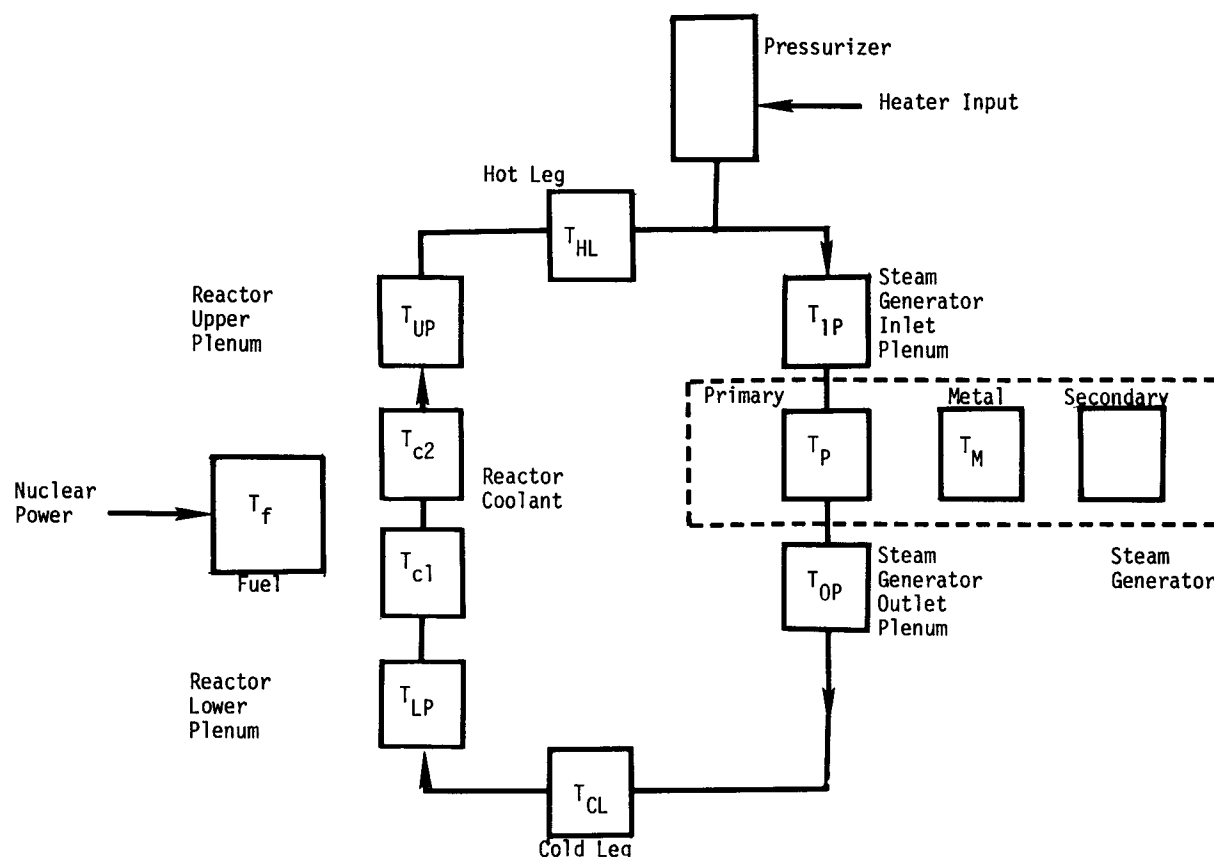


Fig. 6. Nodal structure for complete model.

for all of the state variables is calculated as follows:

$$G(j\omega) = (j\omega I - A)^{-1}h, \quad (53)$$

where

$G(j\omega)$ = vector of frequency response functions

$$j = \sqrt{-1}$$

ω = angular frequency.

IV. EXPERIMENTAL MEASUREMENTS

IV.A. General (Ref. 5)

Experimental data for checking theoretical models can be obtained in the time domain or in the frequency domain. To be useful for model checking, a measurement should have these features:

1. the ability to achieve adequate accuracy while satisfying operational and safety restrictions
2. a sensitivity to changes in the model and its parameters so that achievable experimental accuracy will be adequate to identify model errors.

Both of these factors favor measurements in which many cycles of a low-level periodic input signal are used. Results obtained with simple, nonperiodic signals, such as pulses or steps, can provide data for model checking in the time domain, or these data can be converted into the frequency domain to augment the results obtained with periodic signals. However, tests with nonperiodic signals are usually inadequate for exclusive use because the limited amount of useful signal energy makes it impossible to achieve a high enough signal-to-noise ratio.

Binary (two-level) input signals, or approximations to them, are well suited for input with reactor hardware. Three different binary wave forms have been used in frequency response tests: the pseudorandom binary sequence (PRBS), the n -sequence, and the multifrequency binary sequence (MFBS). Each of these is a binary pulse chain that is characterized by the bit duration and the number of bits in the sequence. A bit is the minimum-width pulse in the chain, and all pulses are an integer multiple of a bit in duration. The frequency response can be obtained by Fourier analysis of the input and output signals at harmonic frequencies of the input signal and by forming the ratio of the Fourier integrals. For a

signal with z bits and a bit duration of Δt sec, the frequencies are $k/(z\Delta t)$ Hz for integer values of k .

The relative amplitudes of the harmonic frequencies are fixed once the number of bits is selected for the PRBS or n -sequence. In both these signals, the shape of the amplitude spectrum is given by

$$\text{amplitude of harmonic } k = C \left| \frac{\sin \frac{k\pi}{z}}{\frac{k\pi}{z}} \right|, \quad (54)$$

where C is a constant and z is the number of bits in the sequence. The PRBS contains all harmonics, and the n -sequence has nonzero harmonic amplitudes only for odd values of k .

The MFBS is more general than the other signals because the user can design a wave form that concentrates the signal energy in frequencies of interest. This is accomplished by a computer optimization of the polarities of the bits so as to minimize the difference between the desired and obtained spectrum for the signal. The advantage of this signal is that it avoids the waste of signal energy in unwanted frequencies. This results in reduced testing time since the signal-to-noise ratio in the desired frequencies is built up faster than for signals with the energy distributed over more frequencies.

The tests used MFBS signals with the specifications shown in Table IV.

IV.B. Procedure for H. B. Robinson Nuclear Plant Tests

Perturbations were made in control rod position and steam valve opening. Both multifrequency binary perturbations and step perturbations were introduced manually by the operator. For the multifrequency binary input, the desired perturbation was prerecorded on a strip chart recorder. It was rewound and then played back for the operator during the test. He performed the necessary operation when a transition on the chart passed under a mark on the window of the recorder. This allowed him to see the desired change and to anticipate an upcoming change.

Though certainly not as satisfactory as an automatic input procedure, this method yielded adequate results.

The control rod perturbations involved a bank of five rods. This bank was divided into two groups, one with three rods and the other with two. Closing a switch on the control panel caused sequential motion of the two groups. The operator was able to manipulate the switch to obtain either one step or two steps when a transition was requested because each step resulted in an audible "click" at the control panel. One series of tests used one-step transitions and the other used two-step transitions. All reactivity variations were <2 cents in amplitude.

Figure 7 shows a portion of the data record for a control rod perturbation test. This shows that the power variation was only $\sim 1.4\%$.

A problem with the control rod position indication was encountered. The indicated control rod position shown in Fig. 7 is obviously erroneous for control rod insertions. The record shows that the neutron flux starts changing before the rod moves, and the rod position indication shows a similar variation on each of these transitions. Since it is not plausible for the magnetic jack control rods to move in this fashion, the apparent rod position was modified for a portion of the record by assuming a step rod position transition at the instant the flux transition began. The corrected rod position signal also appears in Fig. 7. The corrected rod position was determined for one period of data for each test. The frequency response correction factor (actual rod position/measured rod position) was obtained for this period of data and used as a correction on the whole data record. The cause of the incorrect rod position signal has not been determined, but it is suspected that the linear variable differential transformer (LVDT) used for rod position measurement was influenced by the changing local magnetic field that occurred when the control rod drive was actuated.

In the steam flow perturbation test, the operator again introduced the perturbations manually using a prerecorded signal on a strip chart

TABLE IV

Perturbation	Bits in MFBS	Bit Duration (sec)	Number of Periods in Test	Test Duration (min)	Lowest Frequency (rad/sec)	Highest Frequency (rad/sec)
Two-step control rod	256	3	6	76.8	0.00818	0.3027
One-step control rod	128	1	17	36.27	0.0491	1.816
Steam valve	64	20	3	64.0	0.00491	0.1325

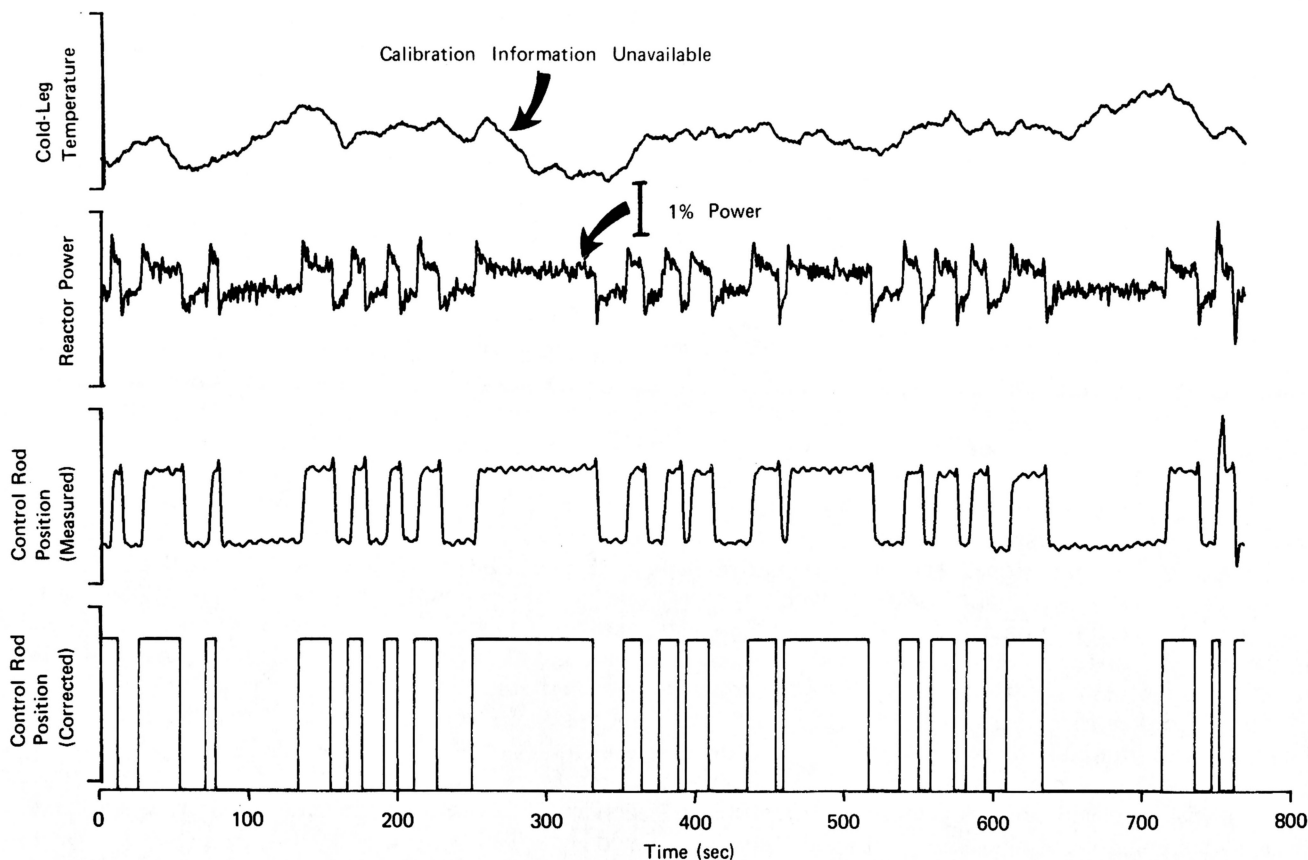


Fig. 7. Experimental responses for reactivity perturbation test.

recorder. Part of the data record from the steam flow perturbation test appears in Fig. 8.

In each test, the following signals were measured:

1. input disturbance signal (control rod position or steam flow)
2. neutron flux
3. hot-leg primary water temperature
4. cold-leg primary water temperature
5. primary loop pressure
6. feedwater flow rate
7. steam pressure
8. core outlet temperature
9. steam generator level
10. steam flow rate.

These signals were conditioned by low-pass filtering and subtraction of the dc bias and were then recorded on an ANSCAN digital data acquisition system.

IV.C. Methods of Analysis

The periodic signal tests were analyzed using a new Fourier transform computer program based on a modification of the Goertzel method.⁵ This method has potential applications for on-line mini-computer analyzers with very small memory requirements (4 to 8K). The use of this method for the H. B. Robinson Nuclear Plant test data analysis provided experience needed to evaluate the method while providing the results for this test. The method was checked by comparisons with results from an earlier computer code⁶ and was found to give correct results.

The computer code that implements the modified Goertzel method is called HFFT. Basically, the method employs a special digital filter that gives a result that can easily be converted into the Fourier transform. The recursion relation for the filter is

$$C(p) = x(p) + \left[2 \cos\left(\frac{2k\pi}{N}\right) \right] \times [C(p-1)] - C(p-2) \quad , \quad (54)$$

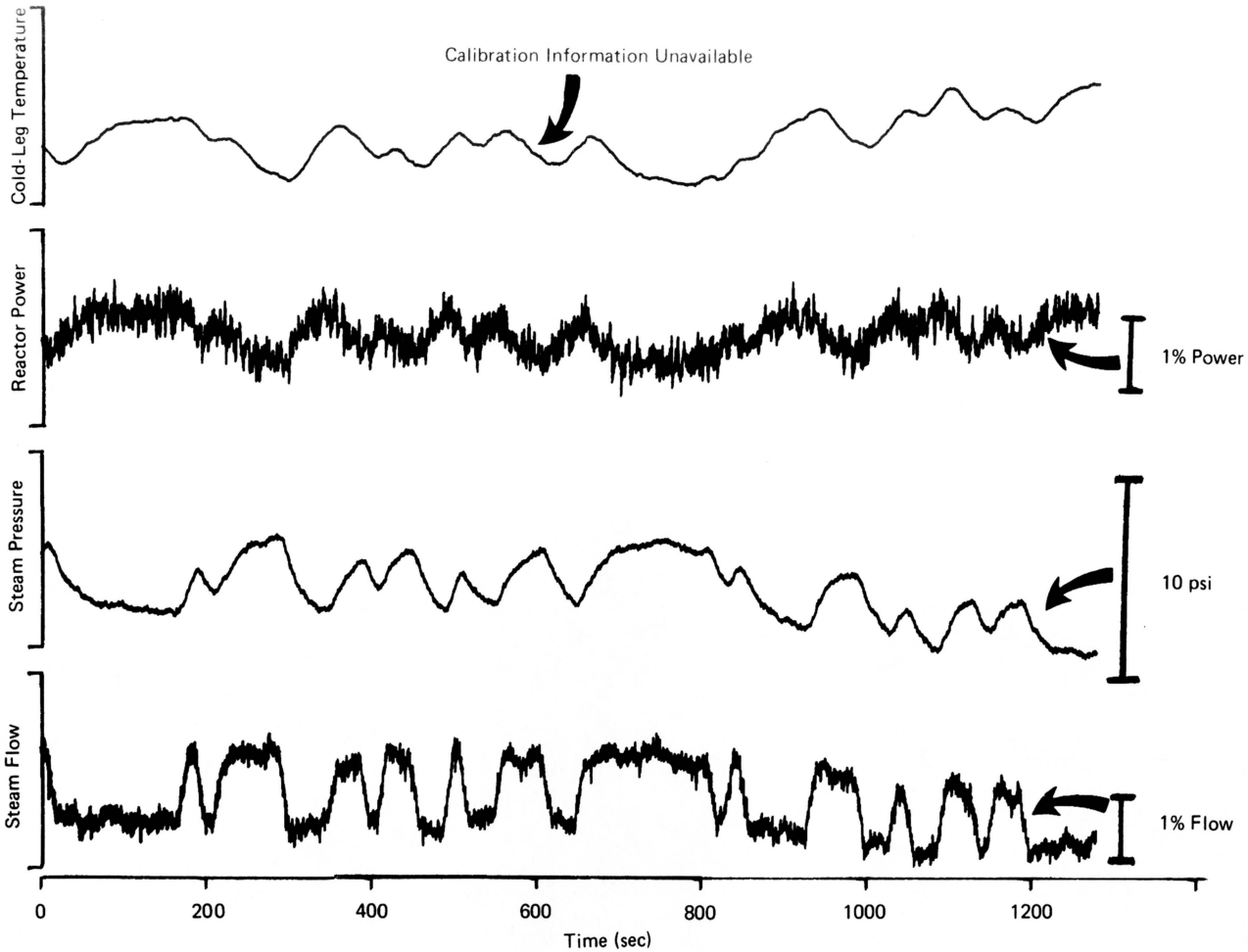


Fig. 8. Experimental responses for steam valve perturbation test.

where

$C(p)$ = p 'th value of the recursion relation

$x(p)$ = p 'th sample of the data record

k = harmonic number

N = number of samples in the data record.

The real and imaginary parts of the Fourier transform are obtained from the last and next-to-last values of the recursion coefficient as follows:

$$Re_k = \frac{1}{N} \left[C(N) \cos\left(\frac{2k\pi}{N}\right) - C(N-1) \right] \quad (55)$$

$$Im_k = \frac{1}{N} \left[C(N) \sin\left(\frac{2k\pi}{N}\right) \right] \quad , \quad (56)$$

where

Re_k = real part of the Fourier integral at the k 'th harmonic frequency

Im_k = imaginary part of the Fourier integral at the k 'th harmonic frequency

$C(N)$ = last value of the recursion coefficient

$C(N-1)$ = next-to-last value of the recursion coefficient.

The frequency response was calculated at the end of each period of data. These results were averaged to give the final estimate of the frequency response.

The coherence function was calculated from the data records and used to assess the quality of the results. The coherence function is defined as follows:

$$\gamma^2 = \frac{|P_{IO}|^2}{P_{II} P_{OO}} \quad , \quad (57)$$

where

P_{IO} = average of cross power spectrum estimates

P_{II} = average of input signal power spectrum estimates

P_{00} = average of output signal power spectrum estimates.

The coherence function has a value of unity when the output signal is perfectly correlated with the input signal (no uncorrelated noise) and a zero value when the output is completely uncorrelated with the input. The proximity of the coherence function to unity is a measure of the quality of the results.

V. RESULTS

V.A. Control Rod Perturbations

Figures 9 through 12 show the frequency response results obtained by analysis of the data records. In each case, an arbitrary decision was made on the minimum coherence points to include. For strong signals, the coherence was high at a number of frequencies, and it was possible to obtain enough high-confidence points to define the frequency response. This was the case with the

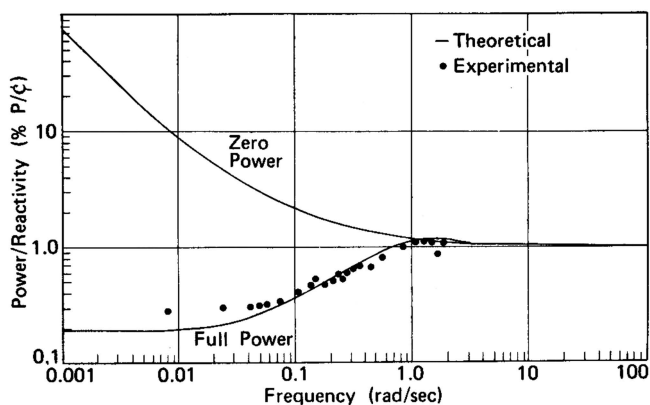


Fig. 9a. Gain of power/reactivity frequency response.

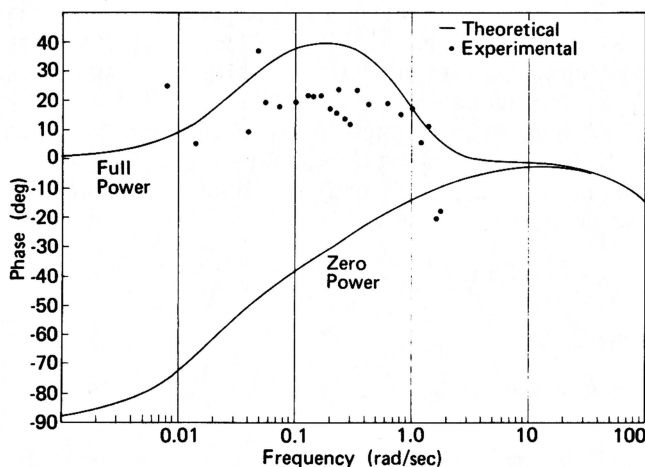


Fig. 9b. Phase of power/reactivity frequency response.

reactor power response where only points with coherence >0.95 are included. For weaker signals, one must use lower coherence results to define the frequency response, but one must remember that these results have a lower confidence level. This was the case with the signals other than reactor power. Here points with a coherence >0.75 were accepted.

V.B. Steam Valve Perturbations

Figures 13, 14, and 15 show the frequency response results for the steam valve perturbation

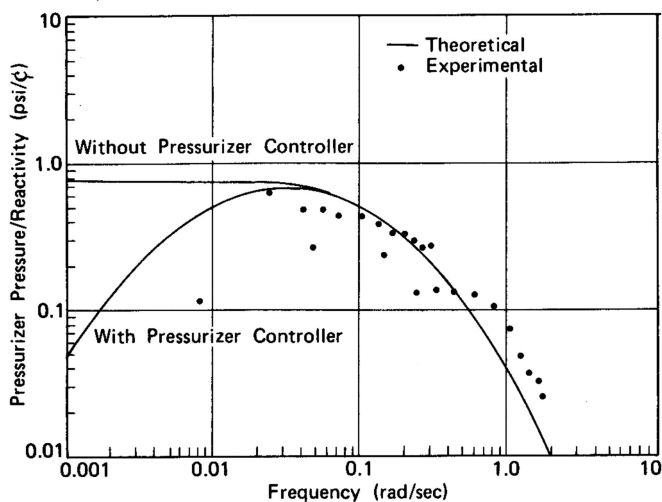


Fig. 10a. Gain of pressurizer pressure/reactivity frequency response.

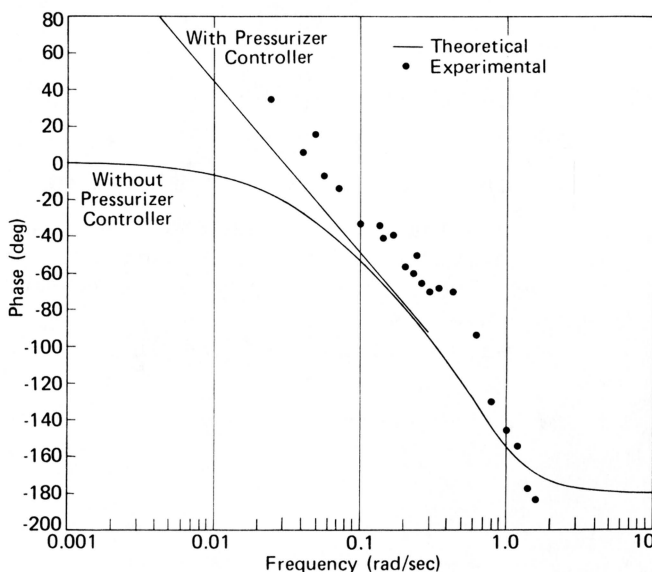


Fig. 10b. Phase of pressurizer pressure/reactivity frequency response.

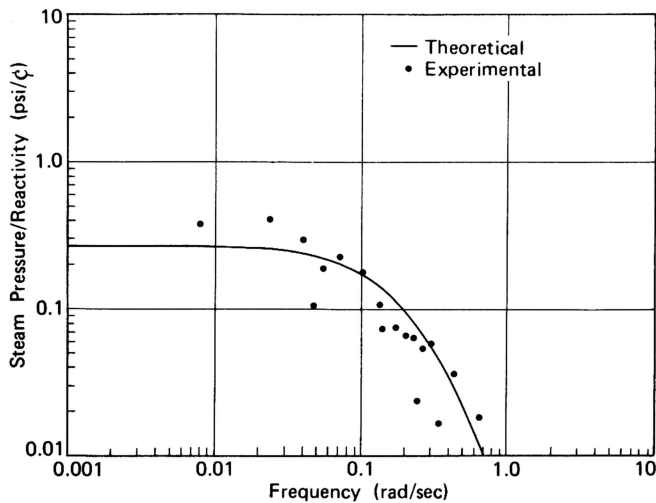


Fig. 11a. Gain of steam pressure/reactivity frequency response.

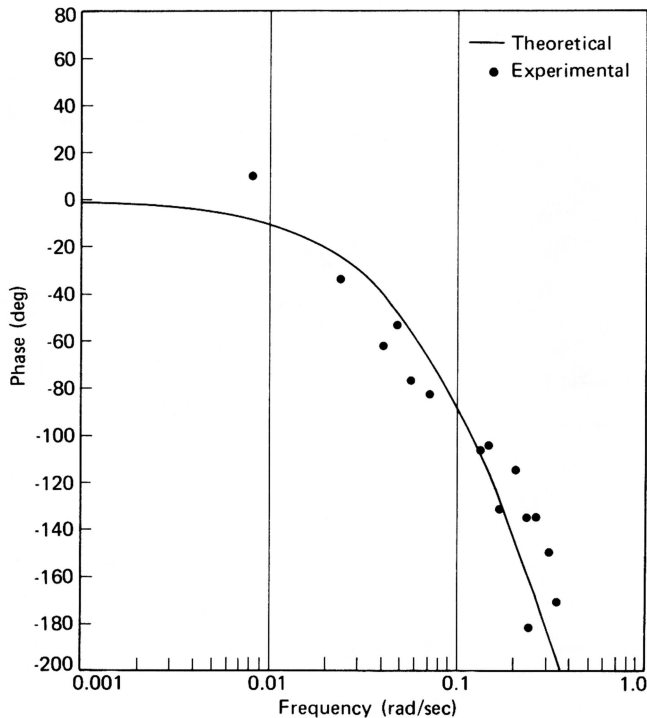


Fig. 11b. Phase of steam pressure/reactivity frequency response.

tests. As with the control rod perturbation tests, a selection was made on the minimum acceptable coherence for each signal. The value used was 0.75.

Figure 16 gives the steam valve step response results; it shows a delay of 10 to 14 sec in the response of the cold-leg temperature when the steam valve is perturbed. The delay is due to the transport time for the primary water as it flows from the steam generator to the cold-leg sensor, which is located in a by-pass line in this plant.

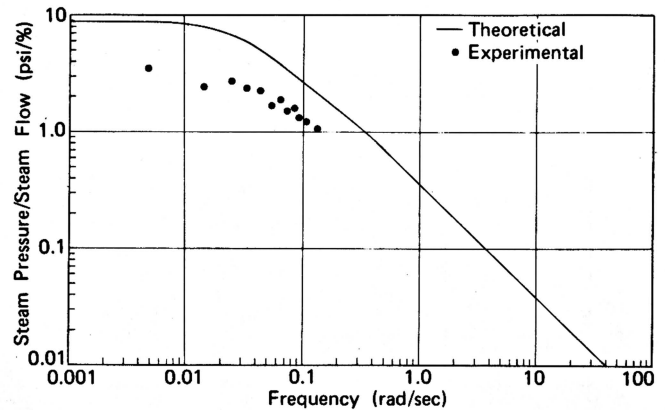


Fig. 12a. Gain of steam pressure/steam flow frequency response.

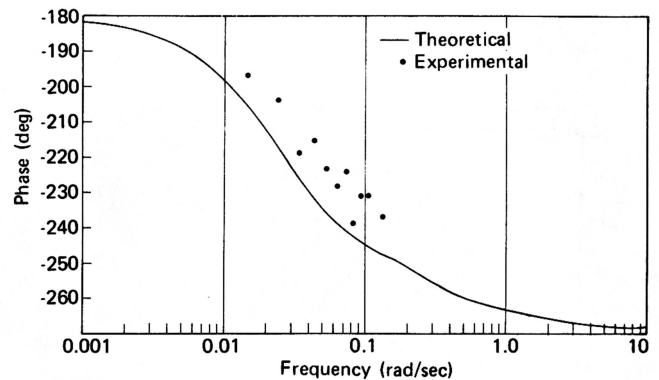


Fig. 12b. Phase of steam pressure/steam flow frequency response.

VI. COMPARISON OF THEORY AND EXPERIMENT

Figures 9 through 15 show theoretical results obtained with the system model as well as the test results. The correlation between theory and experiment for the reactivity perturbation tests (Figs. 9 through 12) is quite good (both in shapes of the curves and absolute values). This demonstrates that the simple model discussed previously is capable of predicting system responses for reactivity perturbations. The model must include the pressurizer controller to predict the primary loop pressure at low frequencies (<0.01 rad/sec). The cold-leg temperature response was not obtained in absolute units because of a data-logging error. Therefore, this response had to be normalized, but the comparison of the curve shapes was good.

The worth of the control rod used for the reactivity perturbation tests was estimated using the test results. As seen in Fig. 9, the theoretical frequency response gain goes to a constant value of 1% power per cent of reactivity change. It

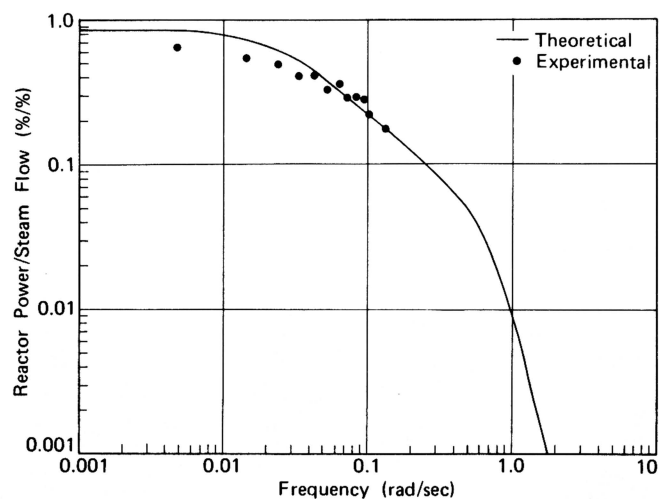


Fig. 13a. Gain of reactor power/steam flow frequency response.

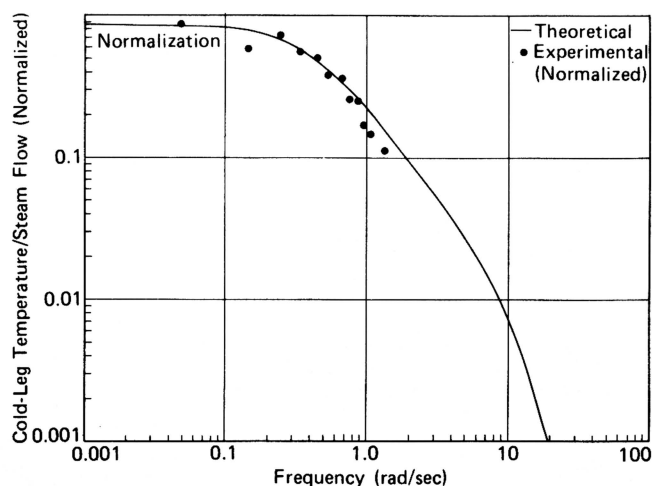


Fig. 14a. Gain of cold-leg temperature/steam flow frequency response.

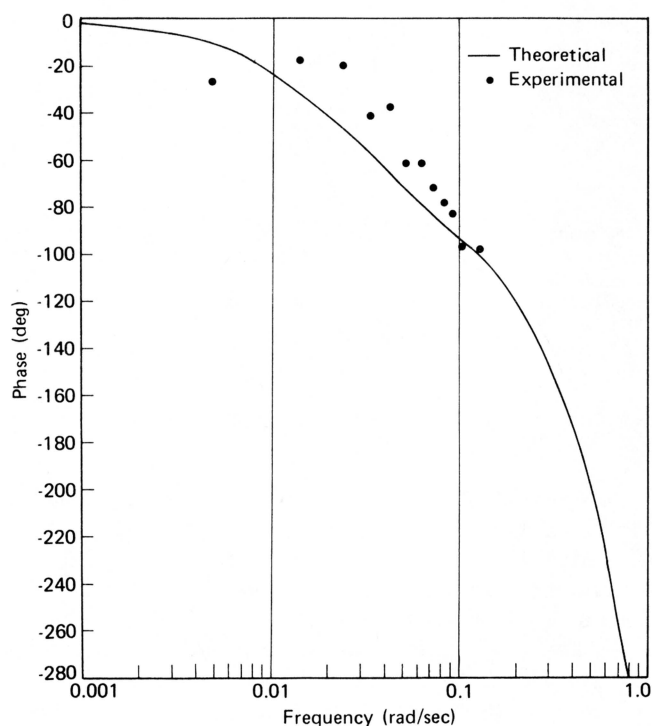


Fig. 13b. Phase of reactor power/steam flow frequency response.

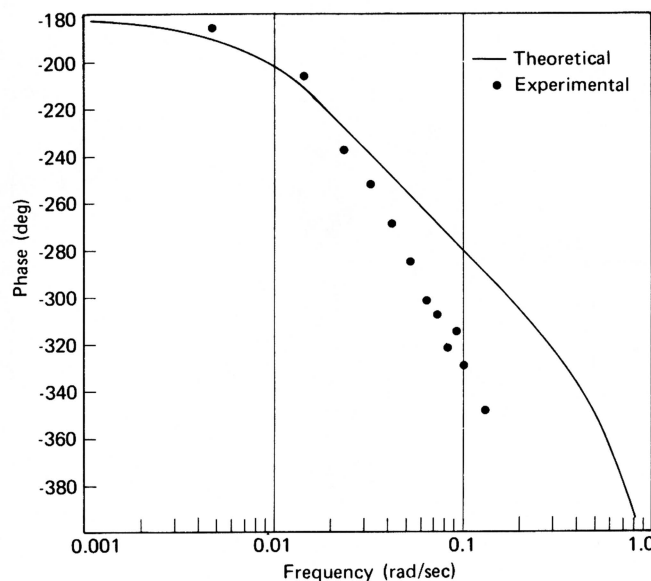


Fig. 14b. Phase of cold-leg temperature/steam flow frequency response.

stays at this value until it approaches the high-frequency break (at 400 rad/sec for the H. B. Robinson Nuclear Plant) where the gain breaks downward. Since there is little uncertainty in the gain in the flat plateau region, results in this frequency range provide a means for evaluating the control rod worth. The measured quantities are the power variations and the control rod posi-

tion variations. The calibration is based on the following relation:

$$\frac{\delta \text{ power } (\%) }{\delta \text{ reactivity } (\text{¢})} = \frac{\delta \text{ power } (\%) }{\delta \text{ rod position (in.)}} \times \frac{\delta \text{ rod position (in.)} }{\delta \text{ reactivity } (\text{¢})} \quad (58)$$

The left side is unity for the frequencies in the plateau region. The first term on the right is the result of the measurement. Therefore,

$$\text{rod worth } (\text{¢/in.}) = \frac{\delta \text{ power } (\%) }{\delta \text{ rod position (in.)}} \quad (59)$$

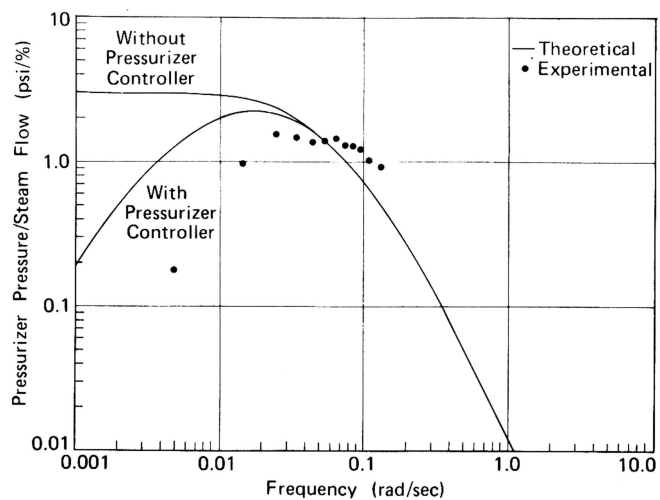


Fig. 15a. Gain of pressurizer pressure/steam flow frequency response.

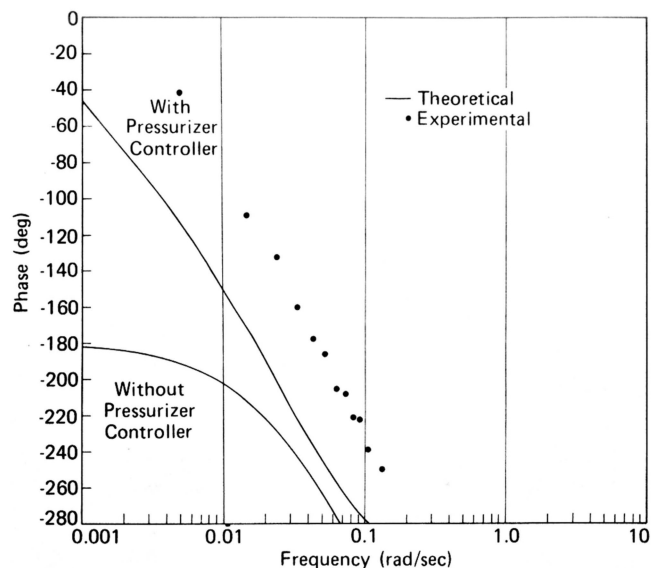


Fig. 15b. Phase of pressurizer pressure/steam flow frequency response.

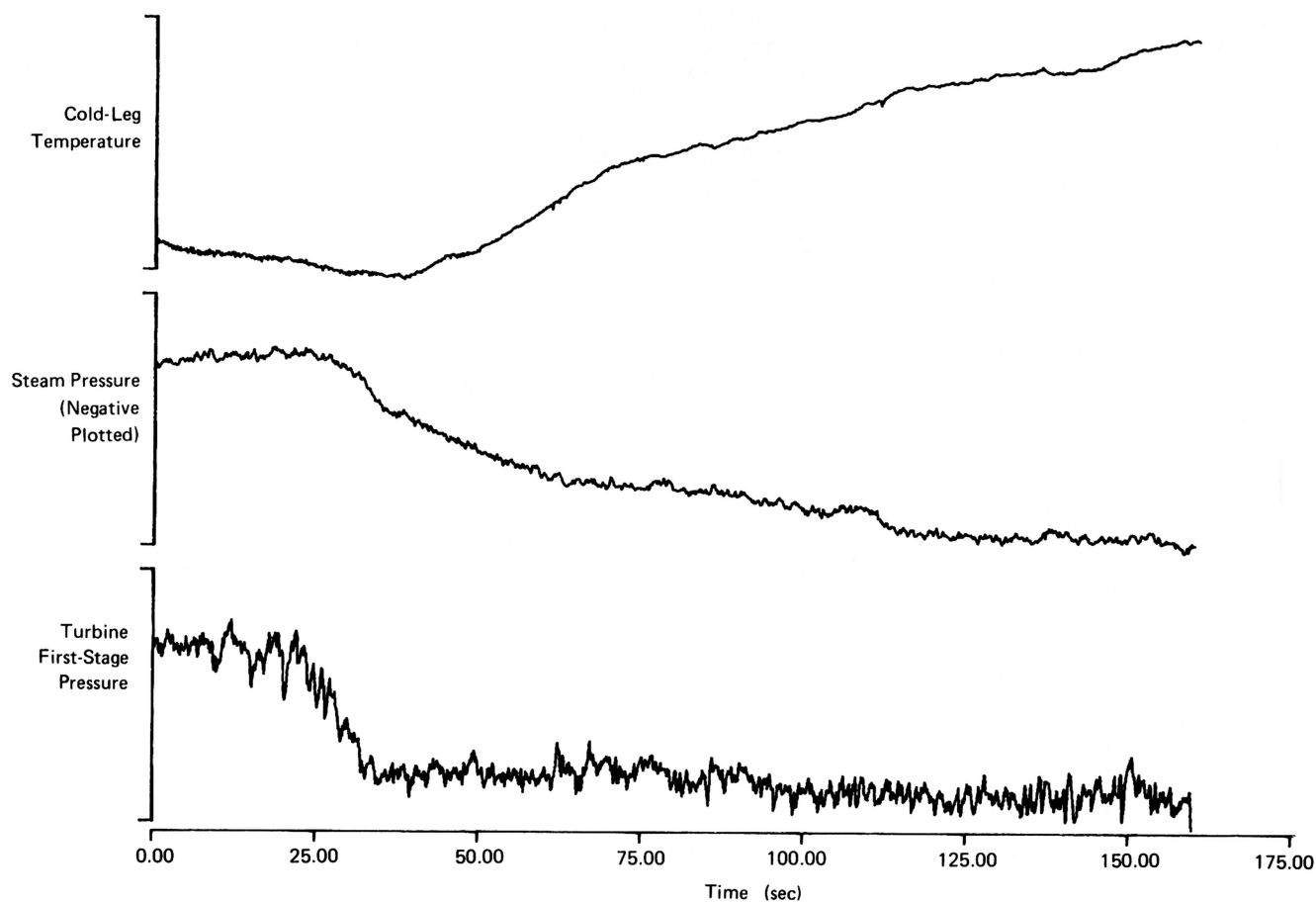


Fig. 16. Responses for a steam valve step test.

This procedure was used to find a rod worth of 1.88¢/in. or 1.17¢/step. (A step involves rod motion of $\frac{5}{8}$ in.)

The correlation of theory and experiment for steam valve perturbation test results was not as good as with the reactivity perturbation tests. This is probably due to the absence of a representation of the steam generator feedwater control system in the system model. The actual feedwater controller uses steam flow, feedwater flow, and downcomer level signals to adjust feedwater flow. To include the feedwater controller, it would be necessary to represent the downcomer level error signal. This would require the addition of an approximate representation in the simplified model or an explicit treatment of downcomer level dynamics in a more detailed model.

VII. CONCLUSION

Small signal perturbations in control rod position and steam valve opening can be introduced manually by the operator of a PWR, and the resulting small fluctuations in plant outputs can be processed to yield data suitable for model validation.

Tests performed at the H. B. Robinson Nuclear Plant, while it operated at full power, provided the following information:

1. Agreement between theory and experiment for four different responses (reactor power, primary pressure, cold-leg temperature, and steam pressure) when control rod position was modulated indicate the adequacy of a simple model for reactivity perturbations. Also, the adequacy of system parameters given in the Safety Analysis Report for specifying the system was demonstrated.

2. The agreement between theory and experiment for steam valve perturbations was not as good, probably indicating the need to add a feedwater control system representation to the plant model to predict responses to perturbations introduced in the steam generator.

3. Control rod calibration can be accomplished by processing the test data if results are obtained above ~1 or 2 rad/sec in a PWR. This provides a simple method for at-power control rod calibration as well as eliminating accurate *a priori* calibration data as a prerequisite for a successful dynamic test program.

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